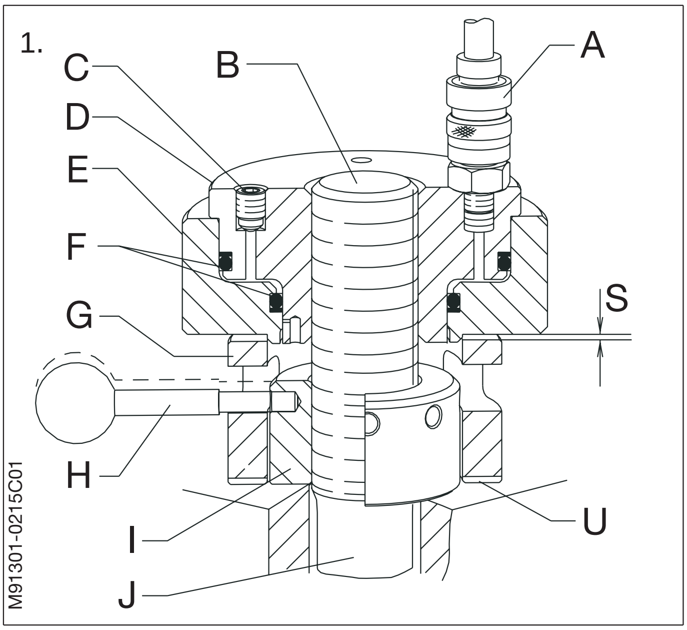
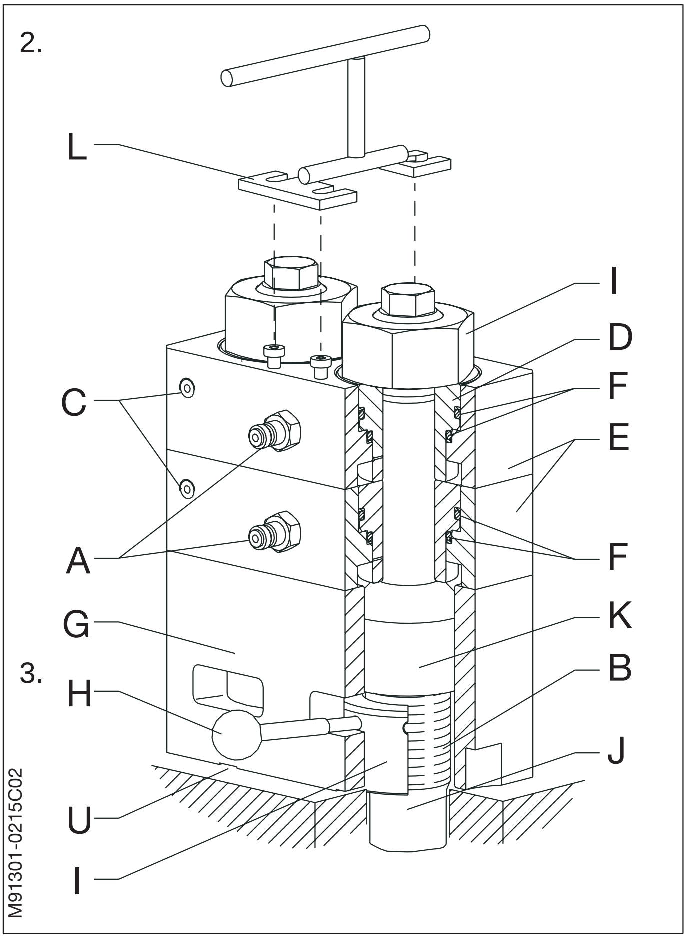
A. Snap-on coupling
B. Tool attachment thread
C. Bleed screw
D. Piston
E. Cylinder
F. Sealing rings
G. Spacer ring
H. Tommy bar
I. Nut
J. Stud or bolt
K. Extension stud
L. Lifting tool
S. Clearance
U. Milled recess for feeler gauge





Studs or bolts provided with threads for attaching hydraulic tools and with circular nuts must only be loosened and tightened up by means of the hydraulic tools supplied.

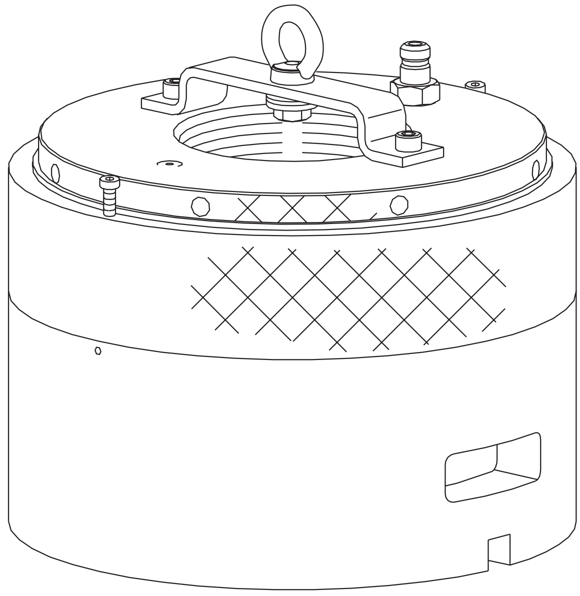
The jack(s) is/are connected, via a distributor block, to a high-pressure pump, which is set to deliver hydraulic oil at the pressure indicated on the jack and on the data sheet in the relevant section of this instruction book. The stud or bolt concerned is thereby lengthened relative to the oil pressure applied and the piston area, and the nut can be loosened or tightened, as required, with the aid of a tommy bar.

The jacks must never be overloaded or exposed to blows or impacts. They are marked with a “Max. lift”, which must not be exceeded.

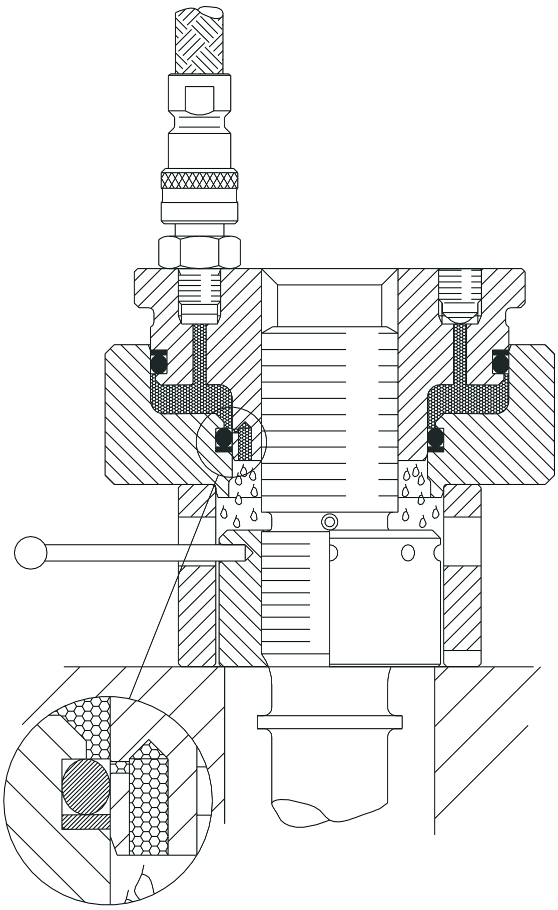
1. The hydraulic tools, except those for the main bearings and the cylinder cover, consist of a jack with an internal thread to suit the tool attachment thread on the stud or bolt, and a spacer ring which is to be placed under the jack and around the nut that is to be loosened or tightened.

2. For the main bearings, the hydraulic tool consists of a double jack with two extension studs. This jack is designed for the simultaneous dismantling of both nuts on one side of the main bearing.

3.



4.



3. For the cylinder cover, the hydraulic tool consists of eight individual jacks suspended from a common lifting tool. A spacer ring is mounted below each jack using two spring pins.

   Operation of the cylinder cover jacks is identical to the operation of the single hydraulic jacks.

4. The hydraulic jacks are so designed that, in the event that the “Max. lift” limit is exceeded, the pressure is relieved at the bottom of the pressure chamber and the oil will be pressed out into the space between the stud and the spacer ring.

   When the pressure is relieved in this way, the lowermost sealing ring will in most cases be damaged. Therefore inspect and, if necessary, replace this sealing ring.

   The oil used must be pure hydraulic oil or turbine oil (with a viscosity of about SAE 20). Oils such as, for instance, lubricating oil (system oil) or cylinder lubricating oil must **not** be used, as these oils are normally alkaline and can thus damage the back-up rings.

   The following instructions must be closely followed to prevent accidents or damage, and after use the jacks should be cleaned and kept in the wooden boxes supplied.

Warning!
Eye protectors and gloves **must be** used when using hydraulic tools.

Single Hydraulic Jack

1. Carefully clean the tool attachment thread, the nut and the surrounding parts.

   Grease the tool attachment thread with molybdenum disulphide grease or with graphite and oil or similar.

   Place the spacer ring around the nut in such a position that the tommy bar can be applied through the slot when the nut is to be loosened.

2. Screw the jack on to the tool attachment thread of the bolt/stud, until the cylinder of the jack bears firmly against the spacer ring.

   Screw back the jack a number of turns, according to the table shown on the sketch, to produce a clearance between the jack and the spacer ring.

   The clearance between the jack and spacer ring ensures dismantling of the jack after loosening the nut.

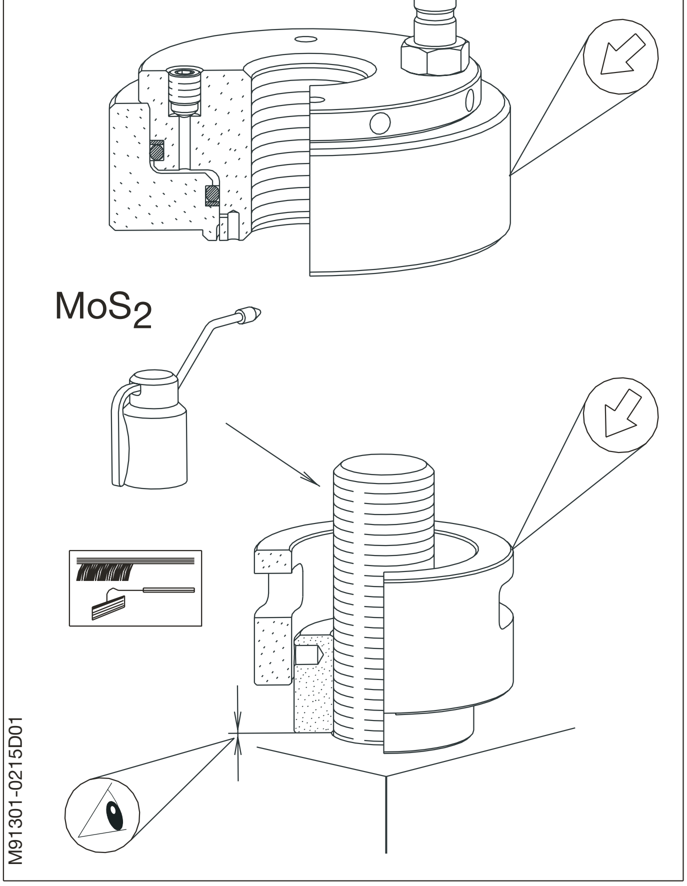
   When loosening nuts on very long bolts (e.g. staybolts) it is recommended to unscrew the jack an additional half turn.

   Connect the hydraulic jack, the distributor block and the high pressure pump by means of high pressure hoses.

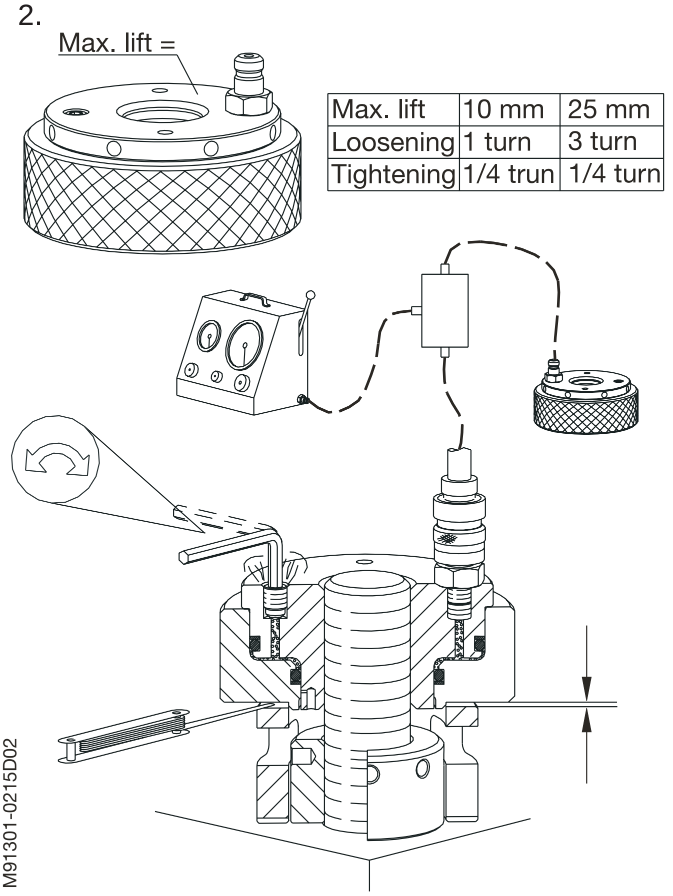
   Loosen the bleed screw in the jack and fill up the system with oil until oil without bubbles, flows out of the bleed hole. Then tighten the bleed screw again.

   After adjustment, check that the parts are guided correctly together and that there is no clearance between the piston and cylinder of the jack.

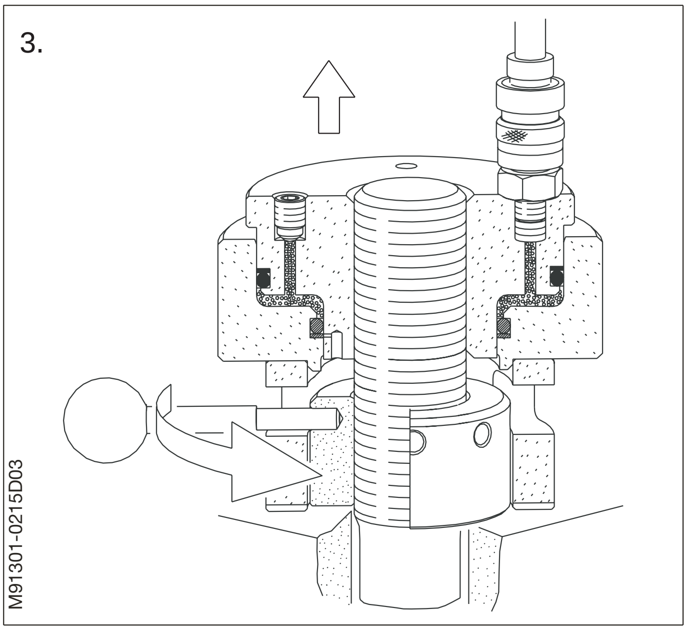
1.

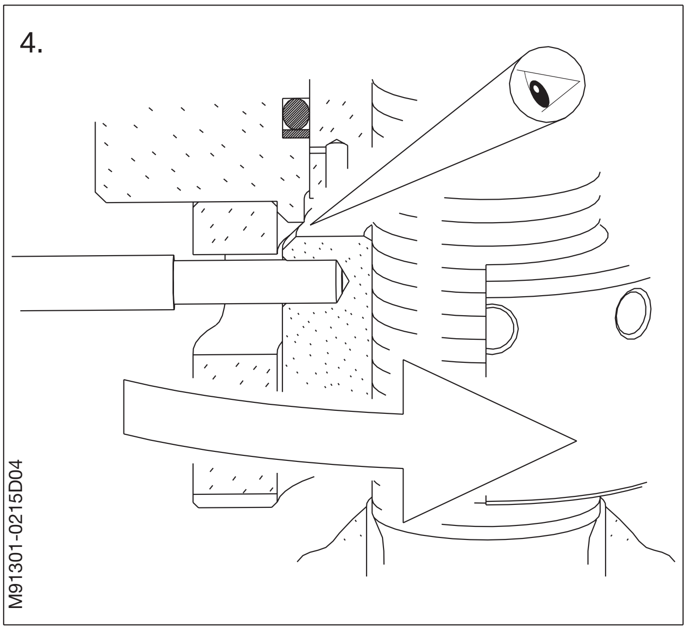


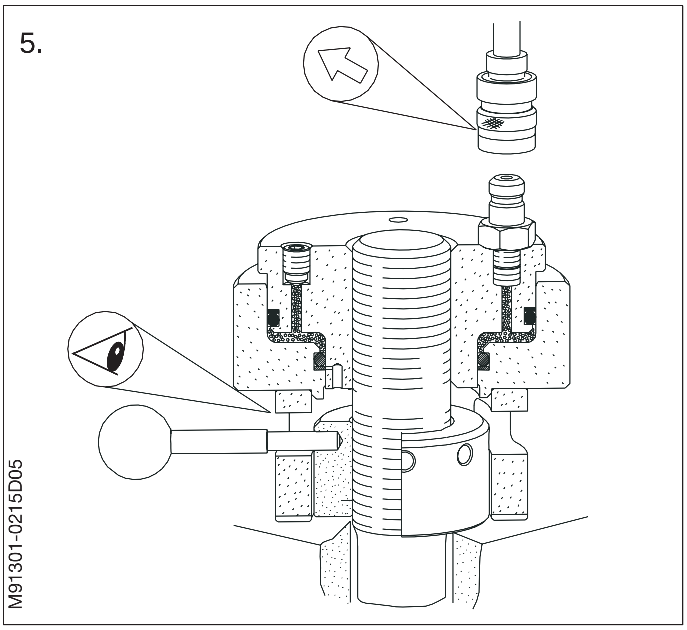
2.



Max. lift	10 mm	25 mm
Loosening	1 turn	3 turn
Tightening	1/4 trun	1/4 turn







3. Increase the oil pressure to the prescribed value. If the nut does not come loose, the pressure may be increased by approx. 150 bar.

4. Unscrew the nut with the tommy bar, making sure that the nut is not screwed up against the jack.

5. Relieve the system of pressure, disconnect the high-pressure pump, and remove the hydraulic tools.

Note!
Make sure **not** to exceed the “max lift” stamped on the jack.

Double Hydraulic Jack

6. Carefully clean the tool attachment threads, the main bearing studs, the nuts and the surrounding parts.

   Grease the tool attachment threads with molybdenum disulphide grease or graphite and oil or similar.

   Mount the extension studs on the main bearing studs.

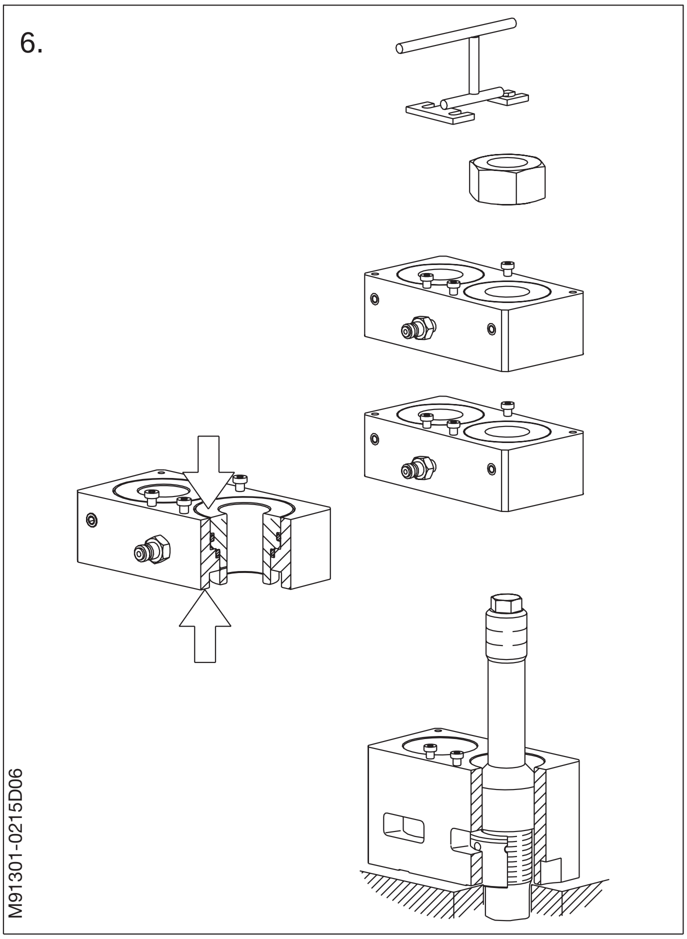
   Using the lifting tool, place the spacer ring over the extension studs in such a position that the tommy bar can be applied through the slots for the purpose of loosening the nuts.

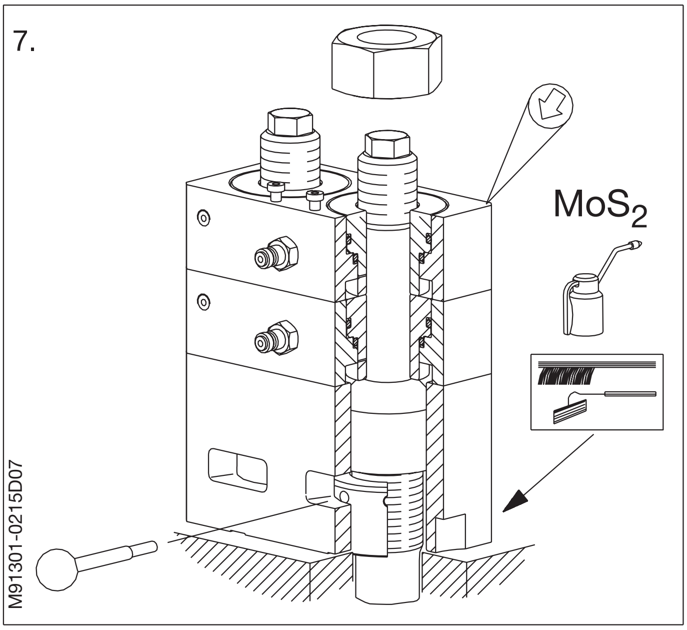
   Press the pistons and the cylinder of both the hydraulic jacks firmly together.

   Mount the two jacks (one by one using the lifting tool) over the extension studs and land them on the spacer ring.

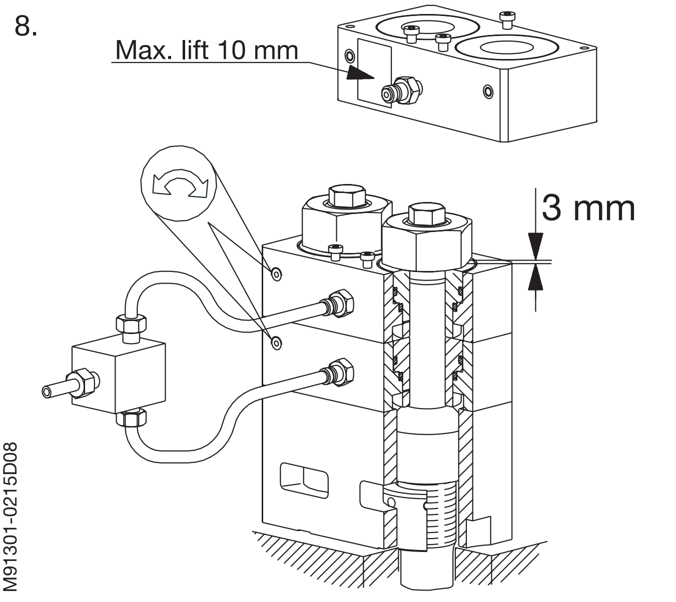
   Make sure that both the cylinders of the jacks and the spacer ring are guided correctly together.

7. Screw the upper nuts on to the threads of the extension studs until the cylinders of the jacks bear firmly against each other and against the spacer ring.

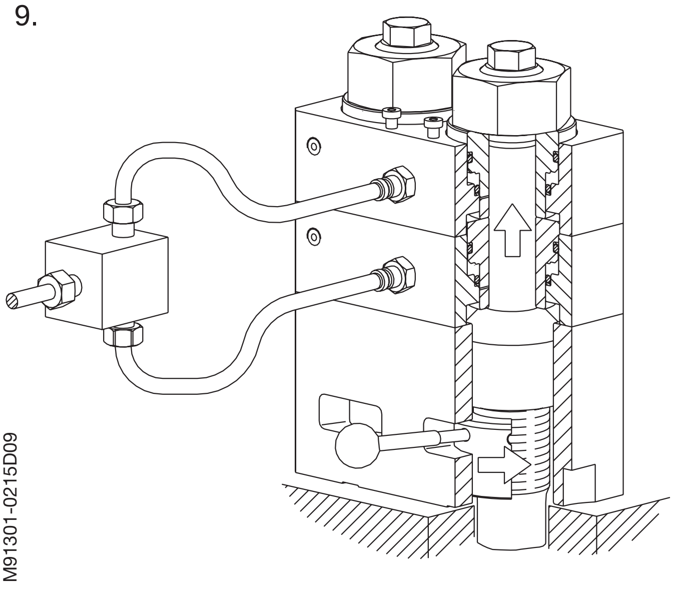




8.



9.



8. The clearance between the pistons and the cylinder of the top jack is adjusted by unscrewing the upper nuts 1 turn ª 3 mm.

   The clearance between the jack and spacer ring ensures dismounting of the jack after loosening the nut.

   Connect the hydraulic jacks, the distributor block and the high pressure pump by means of high pressure hoses.

   Loosen the bleed screws in both jacks, and fill up the system with oil until oil, without bubbles, flows out of the bleed holes.

   Re-tighten the bleed screws.

9. Increase the oil pressure to the prescribed value. If the nut does not come loose, the pressure may be increased by approx. 150 bar.

10. Unscrew the lower nuts 1½ turns with the tommy bar, making sure that the nuts are not screwed up against the extension stud.

11. Relieve the system of pressure, disconnect the high-pressure pump, and remove the hydraulic tools.

Note!
Make sure not to exceed the “max lift” stamped on the jack.

1. The hydraulic jacks require no maintenance except replacement of defective sealing rings, each of which consists of an O-ring and a back-up ring fitted in ring grooves in the piston and cylinder.

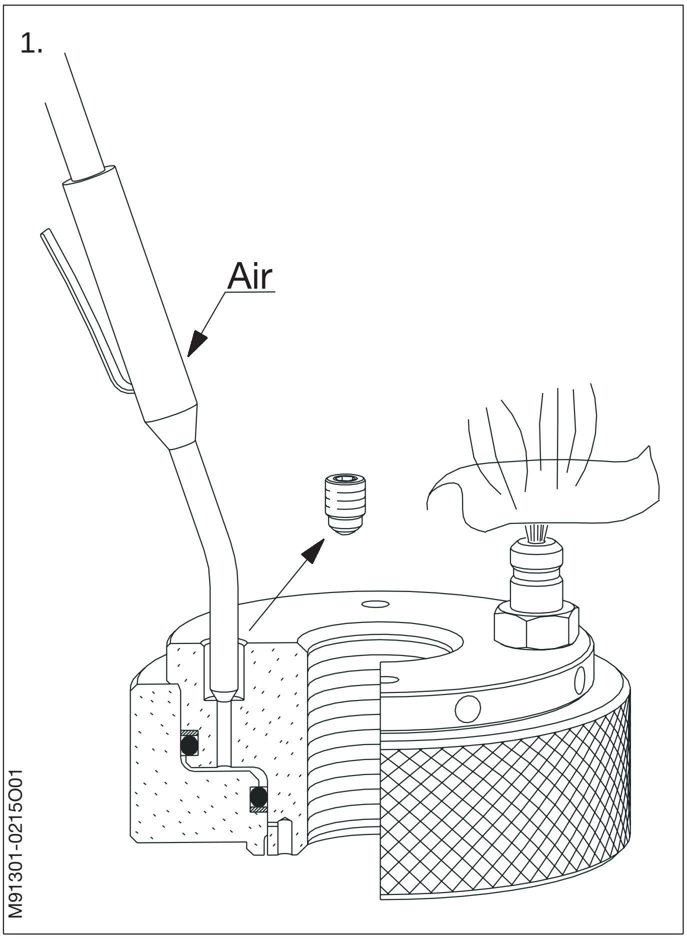
   The piston and cylinder are easily separated by taking out the bleed screw and pressing the parts apart with the help of compressed air.

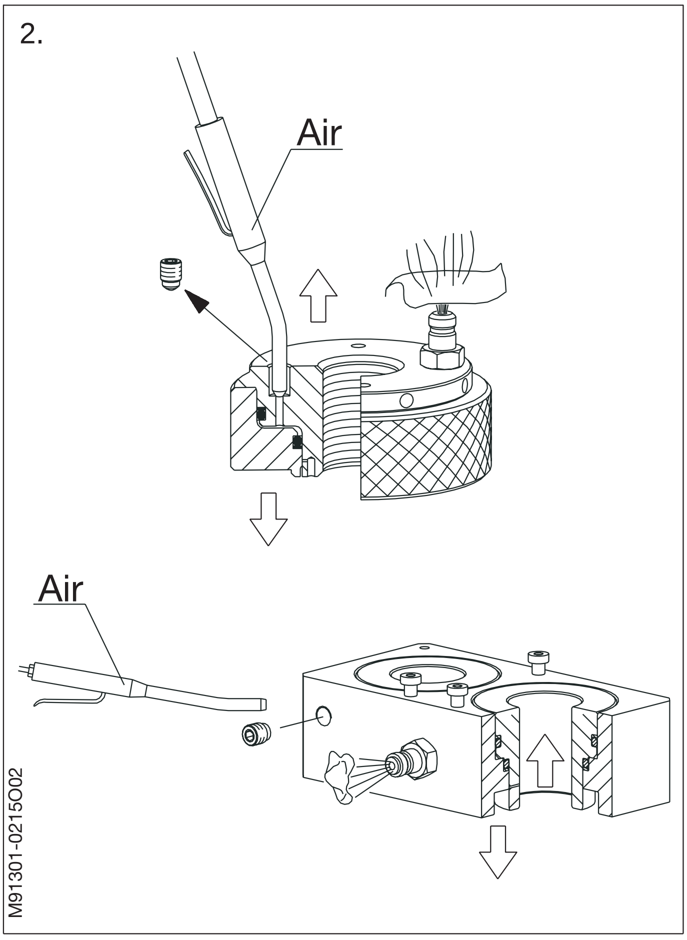
Warning!
Always use protective gloves and eye protection when working with compressed air.

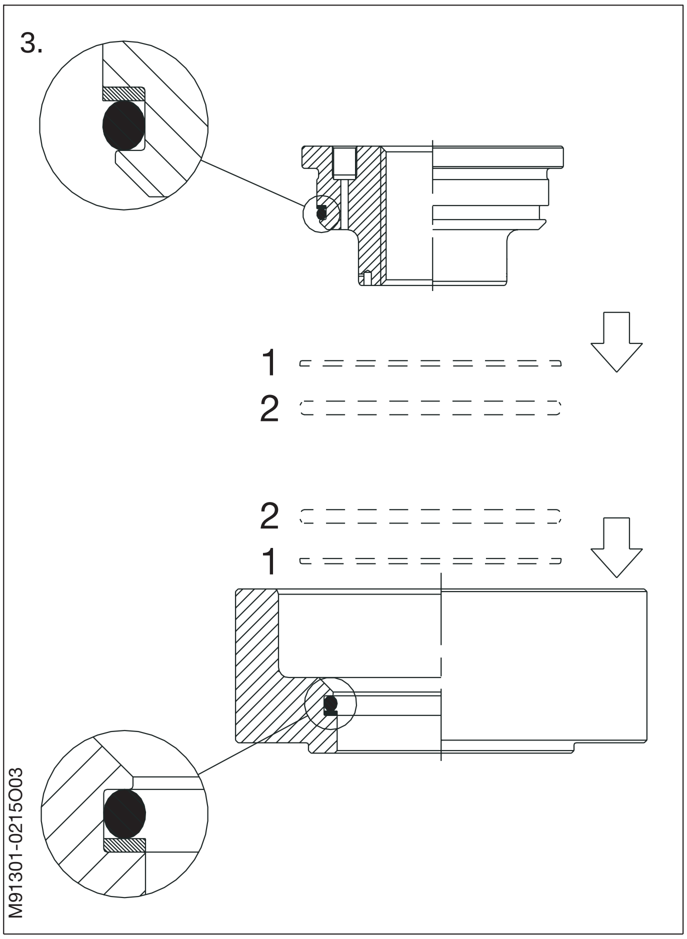
   Make sure that there are no marks or scratches on the sliding surfaces of the parts. The presence of metal particles will damage the sealing rings.

   Keep the sliding surfaces and threads coated with acid-free grease or molybdenum disulphide grease.

2. The pistons and cylinders of the double jacks are separated in the same easy way as described for the single jack.







3. When changing the sealing rings, first mount the back-up ring and then the O-ring.

Note!
Note that the back-up ring (1) **must** be away from the pressure chamber and the O-ring (2) close to the pressure chamber.

See the sketch for the correct mounting of both the upper and the lower sealing rings. The same principle applies to all the jacks.

4. After fitting the sealing rings, coat the piston and cylinder with molybdenum disulphide grease and press the piston and cylinder together. See that the rings do not get stuck between the piston and cylinder.

Single Hydraulic Jack

1. Thoroughly clean the nut, the thread, the contact faces, and the surrounding parts.

   Clean and lubricate the tool attachment thread and the thread in the nut with molybdenum disulphide grease or with graphite and oil or similar.

   Fit the round nut on the thread and tighten it with the tommy bar.

   Check with a feeler gauge that the contact face of the nut bears on the entire circumference.

   Place the spacer ring around the nut in such a position that the tommy bar can be applied through the slot for the purpose of tightening the nut.

2. Press the piston and the cylinder of the jack firmly together.

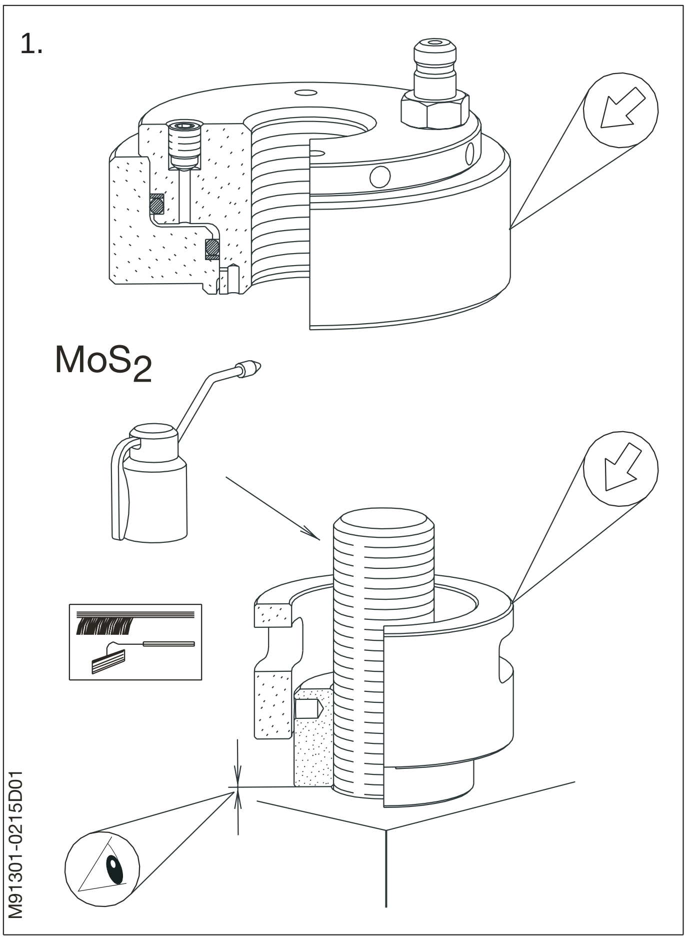
   Screw the hydraulic jack on to the tool attachment thread.

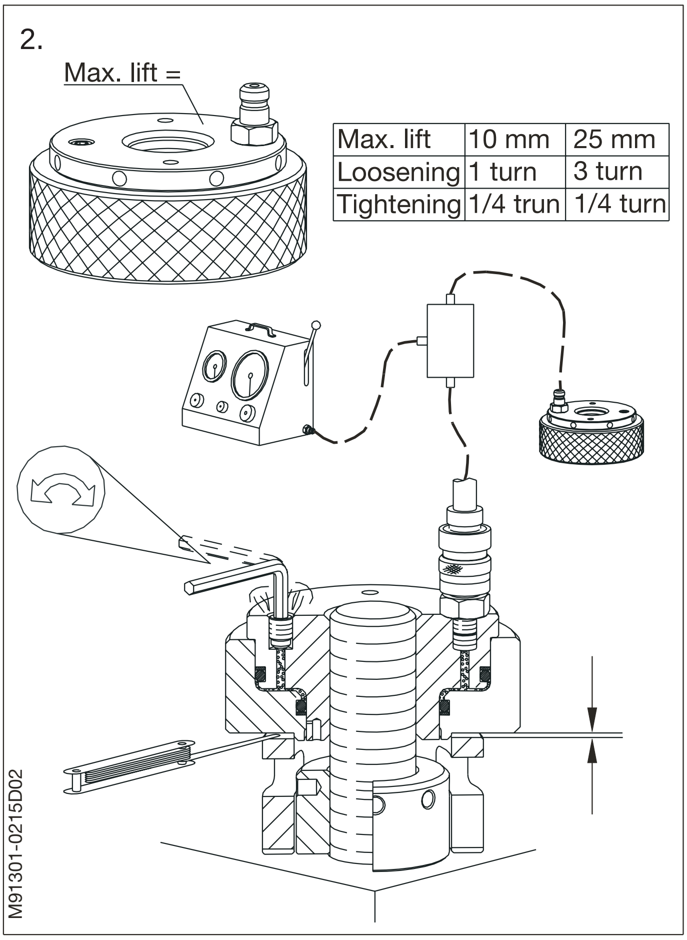
   Make sure that the cylinder of the jack bears firmly against the spacer ring and that the parts are guided correctly together

   Connect the hydraulic jack, the distributor block and the high-pressure pump by means of high-pressure hoses.

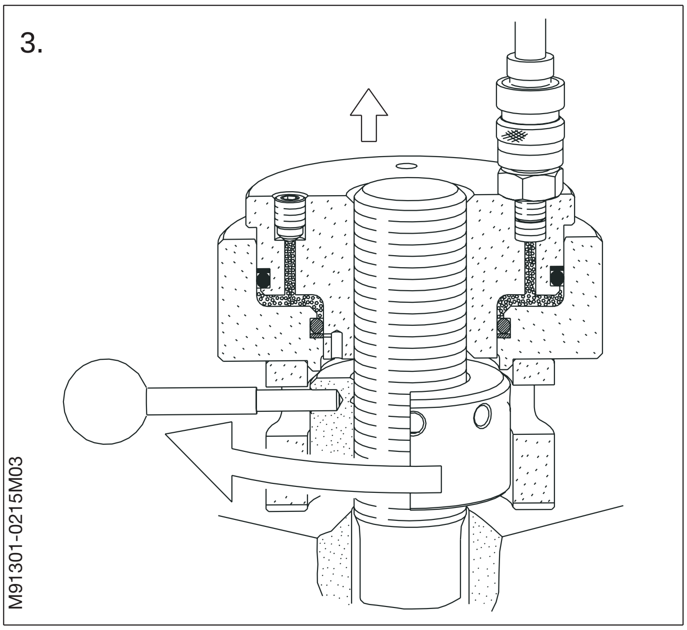
   Loosen the bleed screw in the jack and fill up the system with oil, until oil without bubbles flows out of the bleed hole.

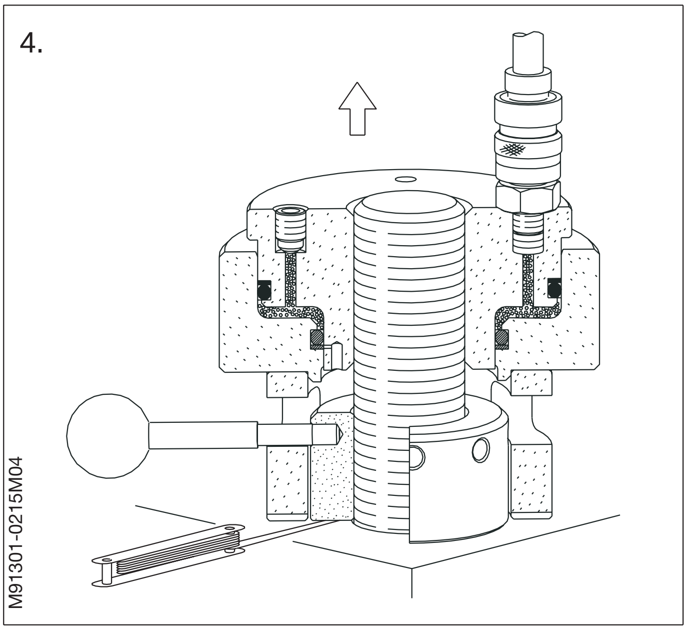
   Re-tighten the bleed screw.





Max. lift	10 mm	25 mm
Loosening	1 turn	3 turn
Tightening	1/4 trun	1/4 turn





3. Increase the oil pressure to the prescribed value, and tighten the nut by means of the tommy bar applied through the slot of the spacer ring.

4. While maintaining the pressure, check with a feeler gauge introduced through the recess at the bottom of the spacer ring that the nut bears against the contact face.

5. Relieve the system of pressure, disconnect the pump, and remove the hydraulic jack.

6. When new studs, bolts or nuts are tightened for the first time, do not remove the jacks, but loosen the nut as described under ‘Loosening’, points 6-7-8, and then tighten the nut again according to the procedure under ‘Tightening', points 17-18-19.

Double Hydraulic Jack

5. Thoroughly clean the nuts, the threads on the main bearing studs, the contact faces, and the surrounding parts.

   Clean and lubricate the tool attachment thread and the threads in the nuts with molybdenum disulphide grease or with graphite and oil.

   Fit the round nuts on the threads and tighten them with the tommy bar.

   Check with a feeler gauge that the contact faces of the nuts bear on the entire circumference.

   Mount the extension studs on the main bearing studs.

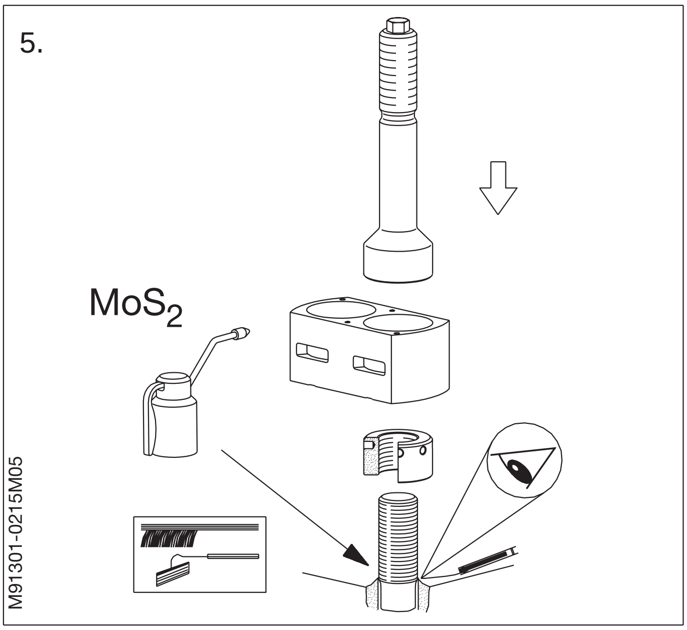
   Using the lifting tool, place the spacer ring over the extension studs in such a position that the tommy bar can be applied through the slots for the purpose of tightening the nuts.

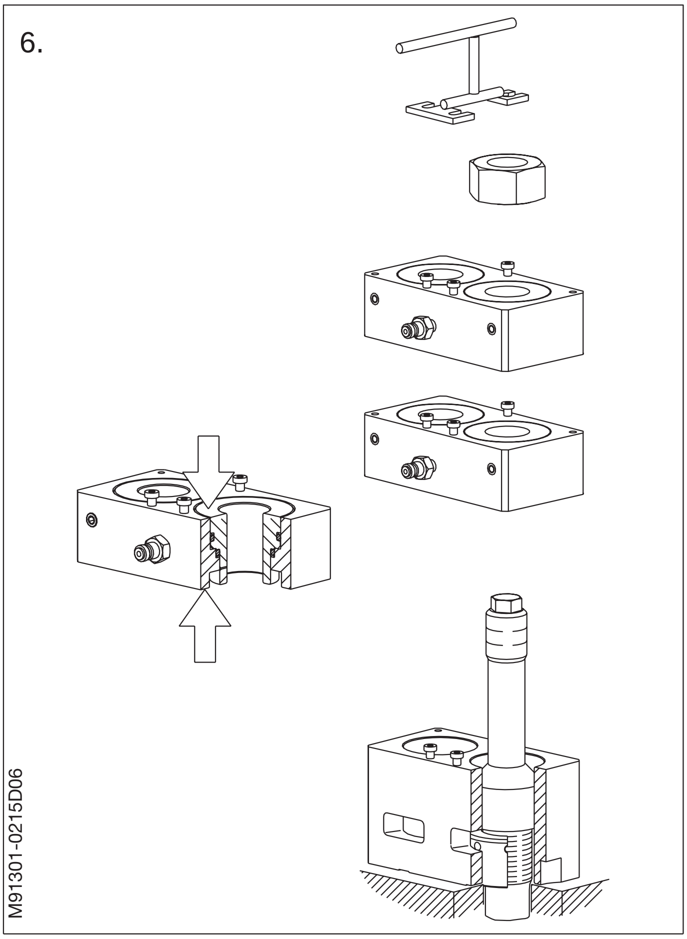
6. Press the piston and the cylinder of both the hydraulic jacks firmly together.

   Mount the two jacks (one by one using the lifting tool) over the extension studs and land them on the spacer ring.

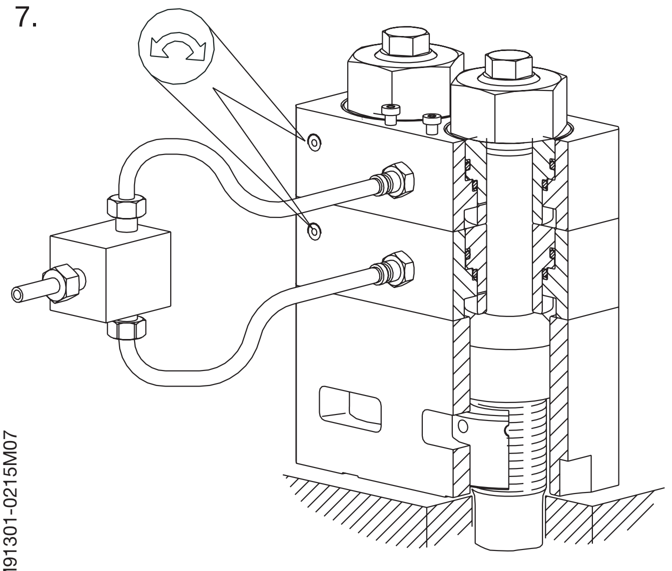
   Make sure that both the cylinders of the jacks and the spacer ring are guided correctly together.

   Screw the upper nuts on to the threads of the extension studs until the cylinders of the jacks bear firmly against each other and against the spacer ring.



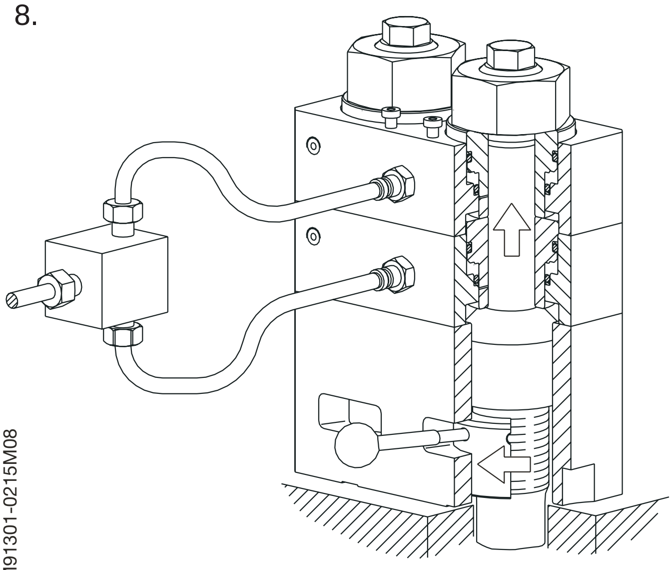


7.



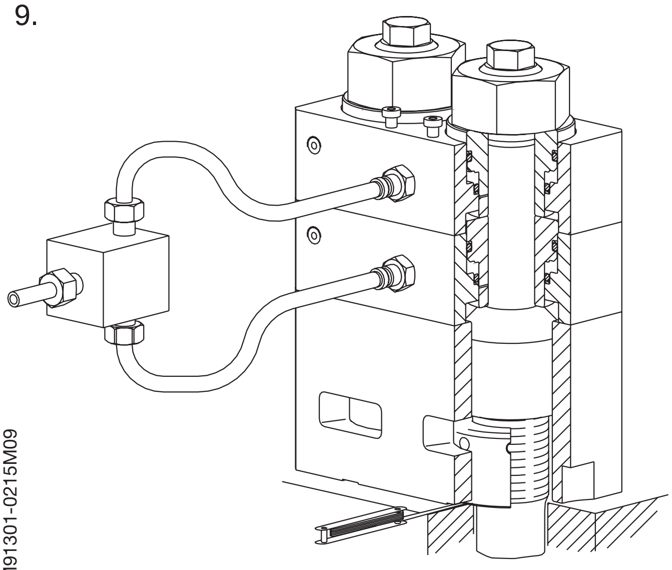
M91301-0215M07

8.



M91301-0215M08

9.



M91301-0215M09

7. Connect the hydraulic jacks, the distributor block and the high pressure pump by means of high pressure hoses.

   Loosen the bleed screws in both jacks, and fill up the system with oil until oil, without bubbles, flows out of the bleed holes.

   Re-tighten the bleed screws.

8. Increase the oil pressure to the prescribed value, and tighten the nuts by means of the tommy bar applied through the slots of the spacer ring.

9. While maintaining the pressure, check with a feeler gauge introduced through the recesses at the bottom of the spacer ring that the nuts bear against the contact face.

10. Relieve the system of pressure, disconnect the pump, and remove the hydraulic tools.

11. When new studs, bolts or nuts are tightened for the first time, do not remove the jacks, but loosen the nuts as described under ‘Loosening', points 12-13-14, and then tighten the nuts again according to the procedure under ‘Tightening', points 24-25-26.

When tightening screws or nuts for which no torque is specified in the instruction procedure or the related data sheet, the standard torques specified in this procedure are to be applied.

1. Standard screws and nuts lubricated with a Molybdenum Disulphide (MoS_2) based lubricant.

Thread	Tightening torque	Thread	Tightening torque
M8	17 Nm	M22	400 Nm
M10	35 Nm	M24	460 Nm
M12	50 Nm	M27	610 Nm
M14	80 Nm	M30	950 Nm
M16	135 Nm	M33	1200 Nm
M18	190 Nm	M36	1650 Nm
M20	260 Nm	M39	2100 Nm

2. Self-locking nuts.

Thread	Tightening torque	Thread	Tightening torque
M8	20 Nm	M22	430 Nm
M10	40 Nm	M24	490 Nm
M12	60 Nm	M27	650 Nm
M14	90 Nm	M30	1000 Nm
M16	150 Nm	M33	1250 Nm
M18	210 Nm	M36	1700 Nm
M20	290 Nm	M39	2200 Nm

3. Screws and nuts locked with glue/Loctite.

Thread	Tightening torque	Thread	Tightening torque
M8	23 Nm	M22	580 Nm
M10	50 Nm	M24	680 Nm
M12	70 Nm	M27	900 Nm
M14	115 Nm	M30	1350 Nm
M16	190 Nm	M33	1700 Nm
M18	270 Nm	M36	2350 Nm
M20	380 Nm	M39	3000 Nm

Before screwing the nuts on, the threads and the contact faces should be greased with a mixture of graphite and oil or molybdenum disulphide with a friction coefficient μ = 0.1-0.12 (e.g. MOLYKOTE paste type G).

The nuts should fit easily on the thread, and it should be checked that they bear on the entire contact face.

In the case of new nuts and studs, the nuts are to be tightened and loosened two or three times so that the thread may assume its definite shape; thus obviating the risk of loose nuts.

Nuts secured with a split pin are to be tightened to the stated torque and then to the next split-pin hole.

The torque spanner must not be used for torques higher than those stamped on it, and it must not be damaged by hammering on it or the like.

Rahsol Torque Spanner

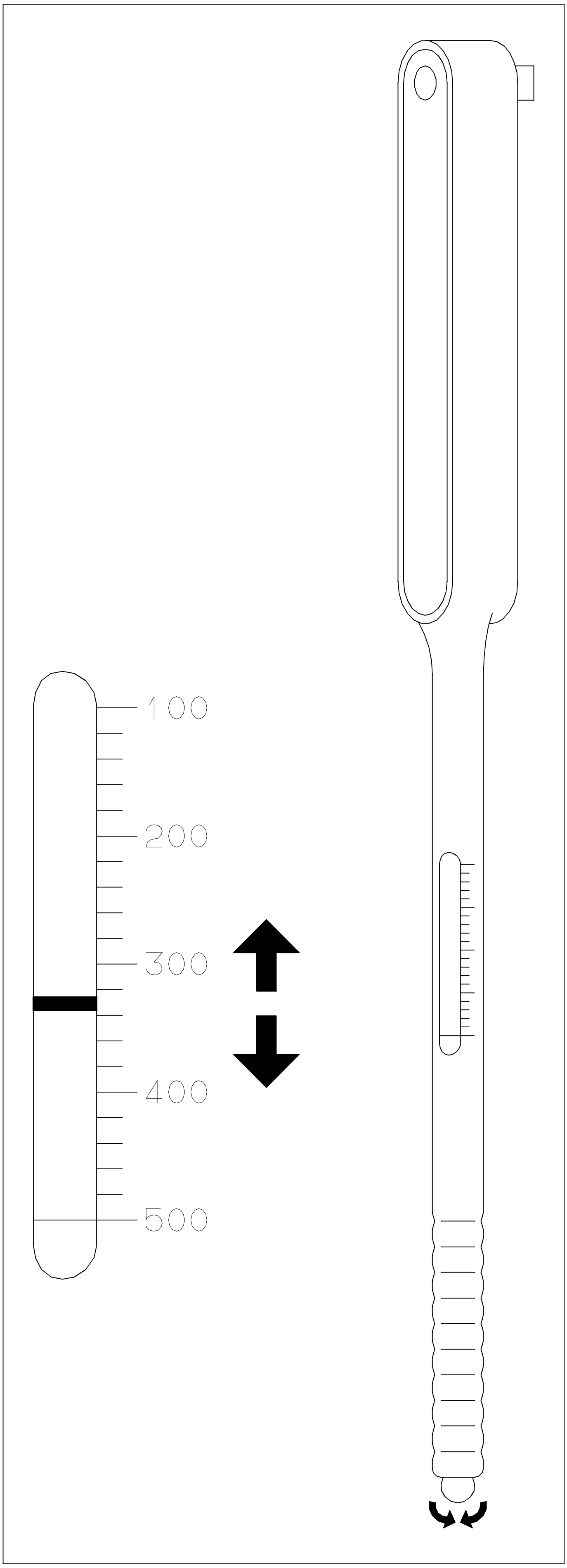
The handle of the torque spanner is provided with a scale indicating the torques at which the spanner can be set.

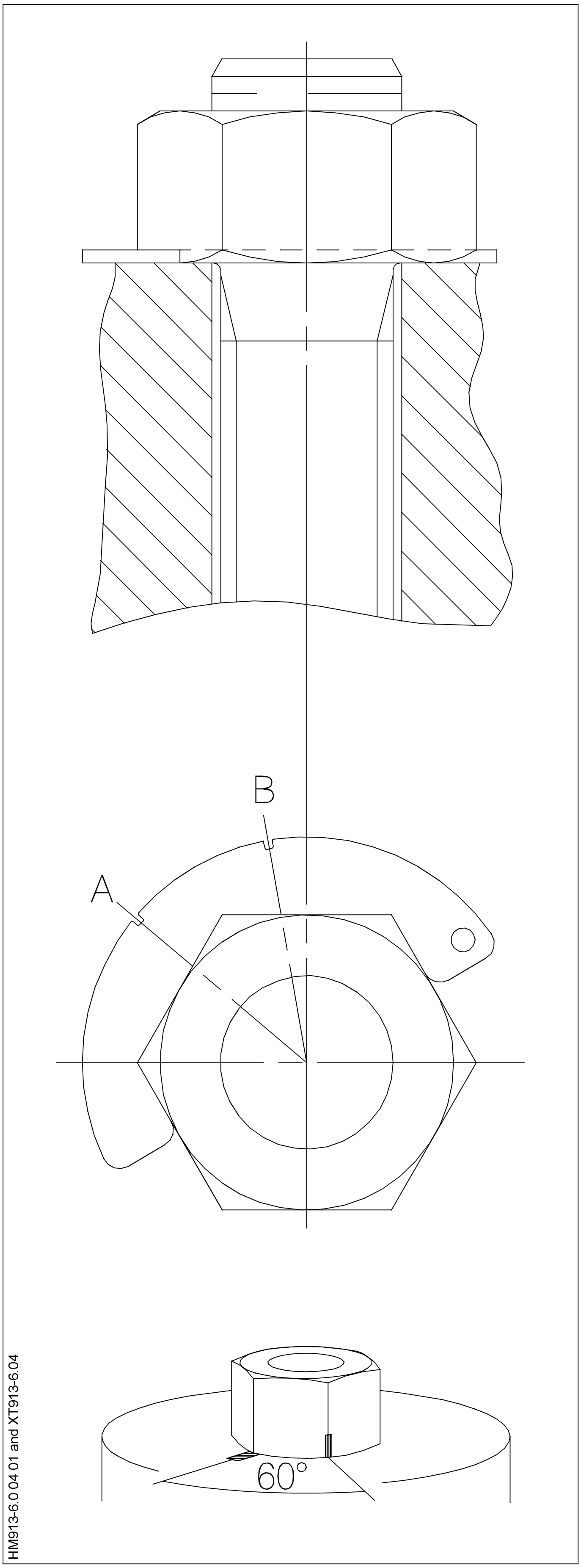
For setting the spanner at the required torque, there is a ball on a small arm at the end of the handle.

When the ball is pulled, a small crank handle appears. A spring-loaded slide in the handle is provided with a mark which, when the crank handle is turned, can be set at the required torque on the scale. The torque spanner works as follows:

The above-mentioned spring acts on a pawl system in the handle. When the pre-set torque has been reached, the pawl system is released, at which moment a small jerk is felt in the spanner, and a small click is heard.

When the torque spanner is not in use, the spring inside should be released by adjusting to minimum torque.





Preparations

Before screwing on the nuts, grease the threads and the contact faces with a mixture of graphite and oil or molybdenum disulphide with a friction coefficient.

μ = 0.1–0.12 (e.g. MOLYKOTE paste type G)

The nuts should fit easily on the thread, and it should be checked that they bear on the entire contact face.

Pre-tightening with a torque spanner

Before tightening the nuts according to a tightening gauge or tightening angle, they must be pre-tightened with a torque spanner.
See Procedure 913-5.

Apply a pre-tightening torque of:
– thread M8 – M20 = 50 Nm
– thread M22 – M27 = 100 Nm
– thread M30 – M39 = 150 Nm
– thread M42 – M48 = 200 Nm

This is in order to ensure a uniform basis for the subsequent tightening with gauge or tightening angle.

Tightening with a tightening gauge

After pre-tightening, place the tightening gauge round the nut, and mark off with chalk on the nut at slot **A** on the tightening gauge, and make another chalk mark on the contact face at slot **B**. Then tighten the nut until the two chalk marks coincide.

Tightening without a tightening gauge

For tightening angles of e.g. 30°–45° and 60°, we usually do not deliver a tightening gauge.
Therefore, after pre-tightening, mark the angle on a corner of the nut and on the contact face, respectively. Then tighten the nut until the two marks coincide.

When tightening new studs or bolts for the first time, loosen again and repeat the procedure – including pre-tightening with a torque spanner, to allow the parts to settle.
See Procedure 913-5.

Some screwed and bolted connections on the engine, as well as some movable joints, are secured against untimely loosening by means of different types of locking devices.

When reassembling the engine after overhauls, it is vital that all such screws and nuts are again locked correctly.

Note!
Make sure only to mount locking devices on nuts and screws which have been mounted with locking devices by the engine manufacturer.

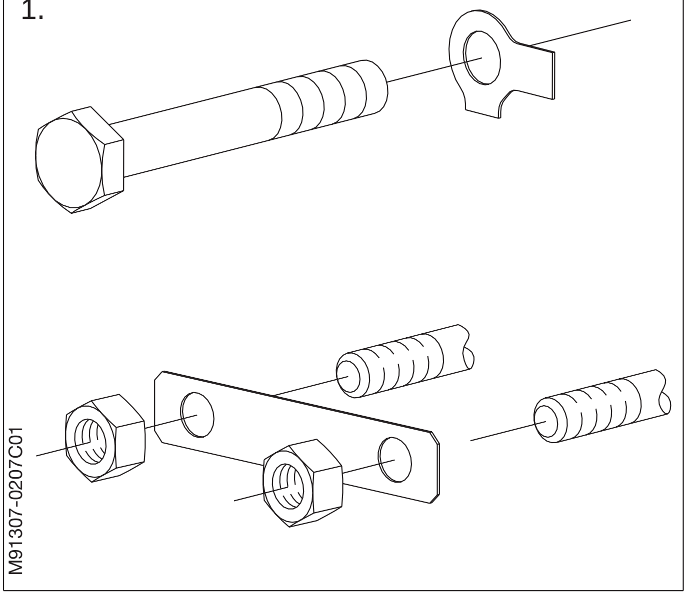
1. **Lock washers**, tab washers, locking plates, etc., must always be replaced. The tab-like projections, etc., on the washers are to be bent back over one of the flats of the screw or nut concerned.

2. Used **spring washers** must be replaced.

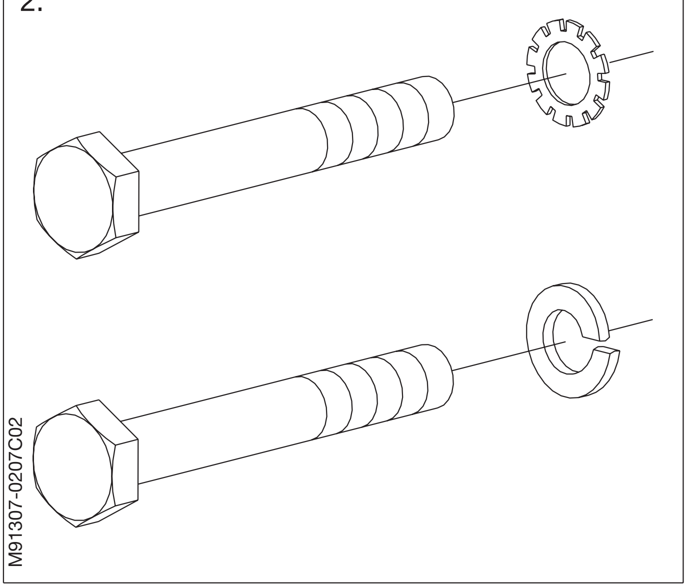
3. **Self-locking nuts** may only be used five times. Therefore, give the nut a centre punch mark each time it is loosened.

   To tighten a self-locking nut to a specific torque, first measure the torque required to turn the nut itself, and then add this torque to the torque value stated on the data sheet of a procedure.

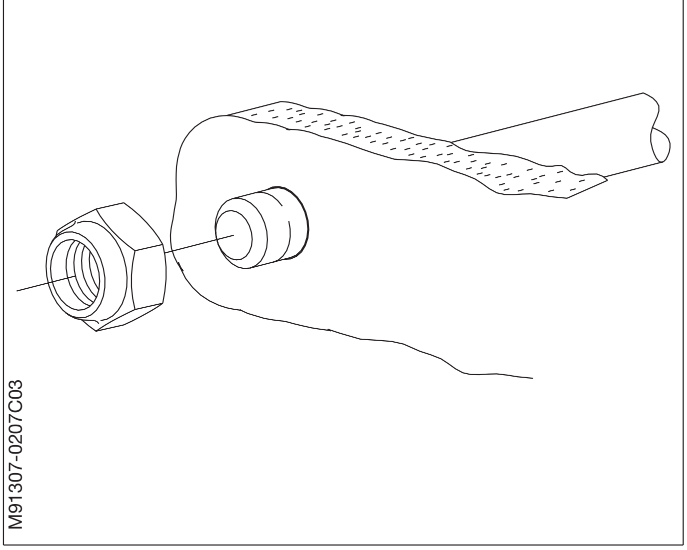
1.

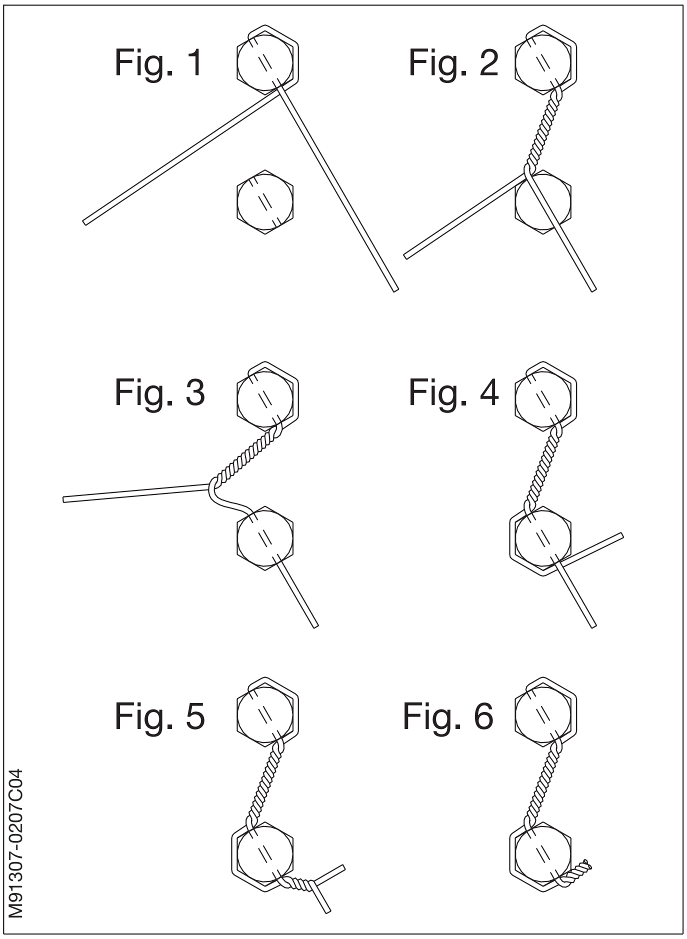


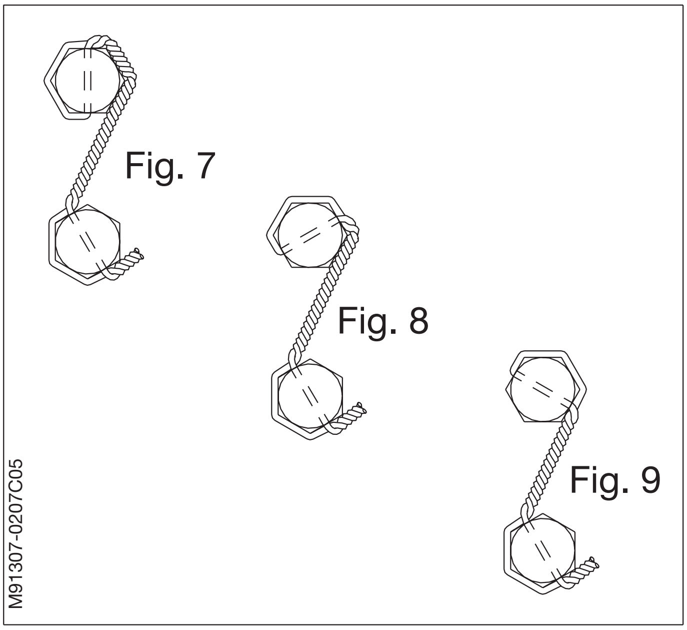
2.



3.







Locking wire should be fitted after the screws or nuts have been tightened to the correct torque. Do not overtighten or loosen the units to get a correct alignment of the wire holes. Always fit new wire after tightening-up the units.

Any tendency of the screws or nuts to loosen will be counteracted by a tightening of the locking wire. Do not secure more than four units in a series, unless otherwise specified.

Fig. 1: Insert wire, grasp the upper end of the wire and bend it around the head of the screw, then under the other end of the wire, be sure that the wire is tight around the head.

Fig. 2: Twist the wire **clockwise** until it is just short of the hole in the second screw. Keeping the wire under tension, twist it until tight. When the wire is tight, the wire shall have approximately 7-10 twists per 25 mm. One twist is a twist of the wires through an arc of 180°, equal to half of a complete turn.

Fig. 3: Insert the uppermost wire in the second screw, and pull it tight.

Fig. 4: Bend the lower wire around the screw, and under the end protruding from the screw.

Fig. 5: Keeping the wire under tension, twist it min. 3 twists, **counterclockwise** until tight.

Fig. 6: During the final twisting motion, bend the wire along the screw head. Cut off excess wire.

Fig. 7-9: Show the preferred ways of mounting the locking wire on screws with wire holes oriented in different angles.

Fig. 10: Shows how to route the locking wire on screws in different planes.

Fig. 11: Wire that passes over the top of a nut is an acceptable alternative only if it routes around the protruding screw thread.

Fig. 12: Wire that passes over the top of a nut is also an acceptable alternative if the hole is located as shown in the figure.

Fig. 13: Where drilling of locking wire holes has caused a thin wall section, route the wire as shown in the figure, to prevent damage to the nut.

Fig. 14: Locking wire can be mounted to any other part of the assembly if nothing else is possible.

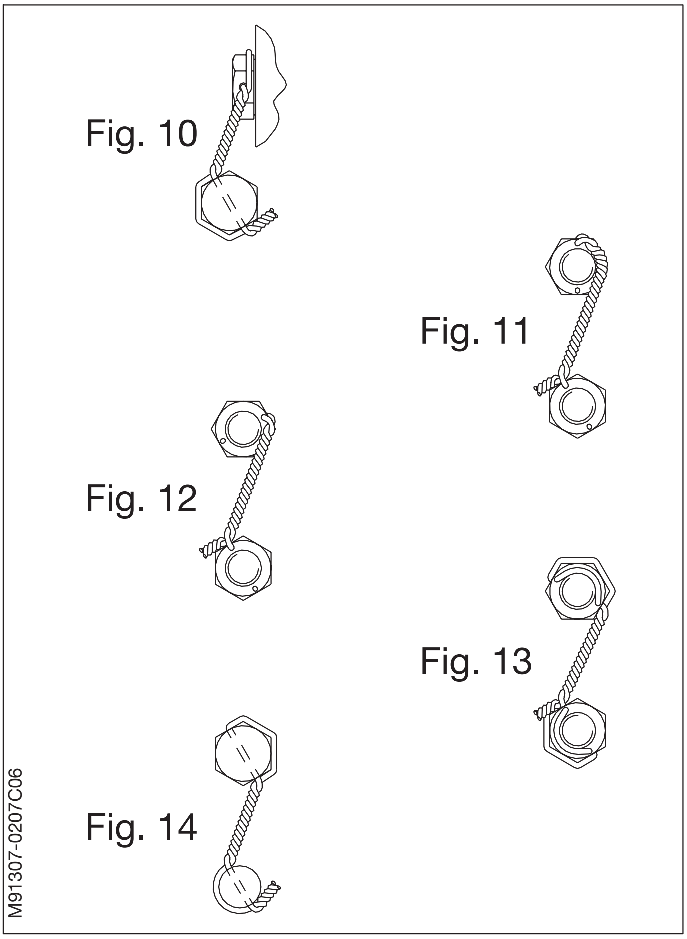
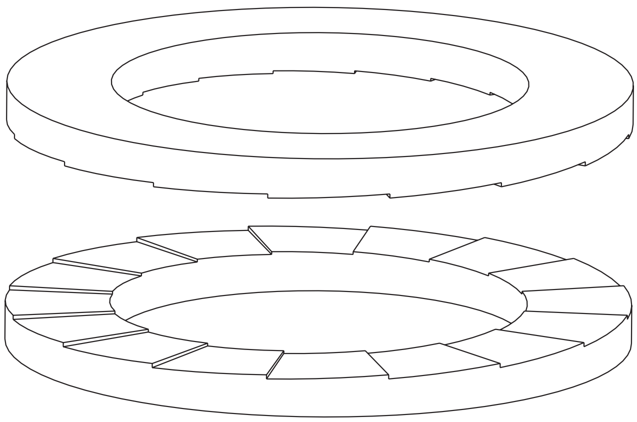
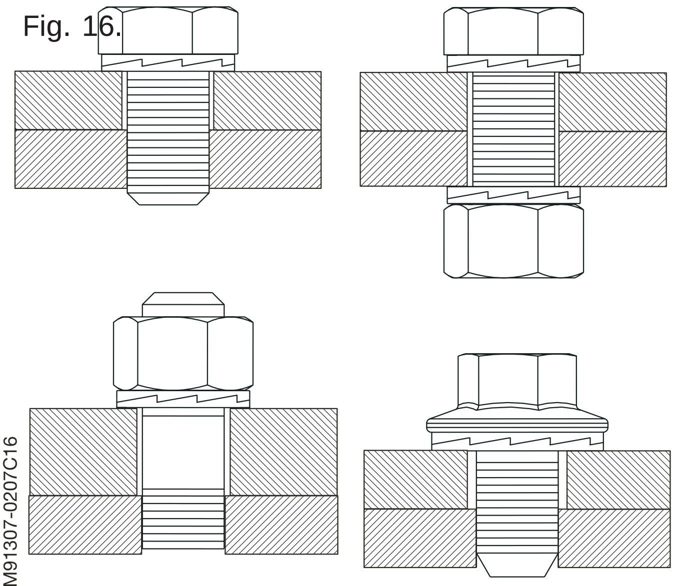


Fig. 15



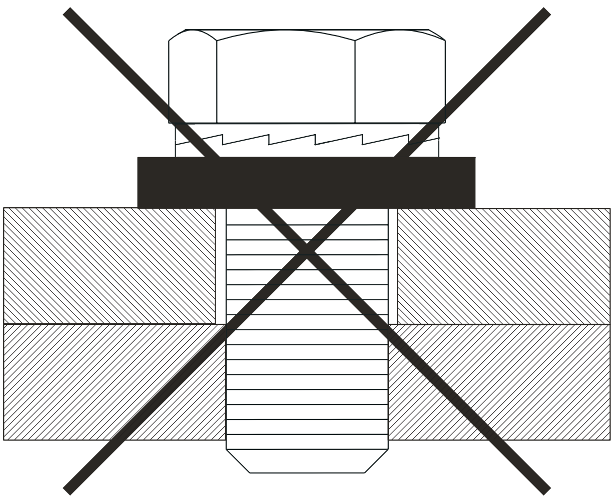
M91307-0207C15

Fig. 16.



M91307-0207C16

Fig. 17



M91307-0207C17

Cam Lock Washers with rising cams on one side and radial teeth on the other.

Fig. 15: The washers are installed in pairs, cam face to cam face. When the bolt and/or nut is tightened the teeth grip and seat the mating surfaces. The cam lock washer is locked in place, allowing movement only across the face of the cams. Any attempt from the bolt/nut to rotate loose is blocked by the wedge effect og the cams.

Fig. 16: Different uses of the cam lock washers.

Fig. 17: Don not use the cam lock washers in conjunction with other loose washers.

Note!
Cam lock washers must be installed in pairs and are not to be substituted by other typers of washers.

Molybdenum Disulphide (MoS_2)

The following procedure is to be followed prior to the mounting of metal surfaced parts which are to function as seals.

- Clean the surface with a cleaning fluid and ensure that the entire surface is completely free of grease.

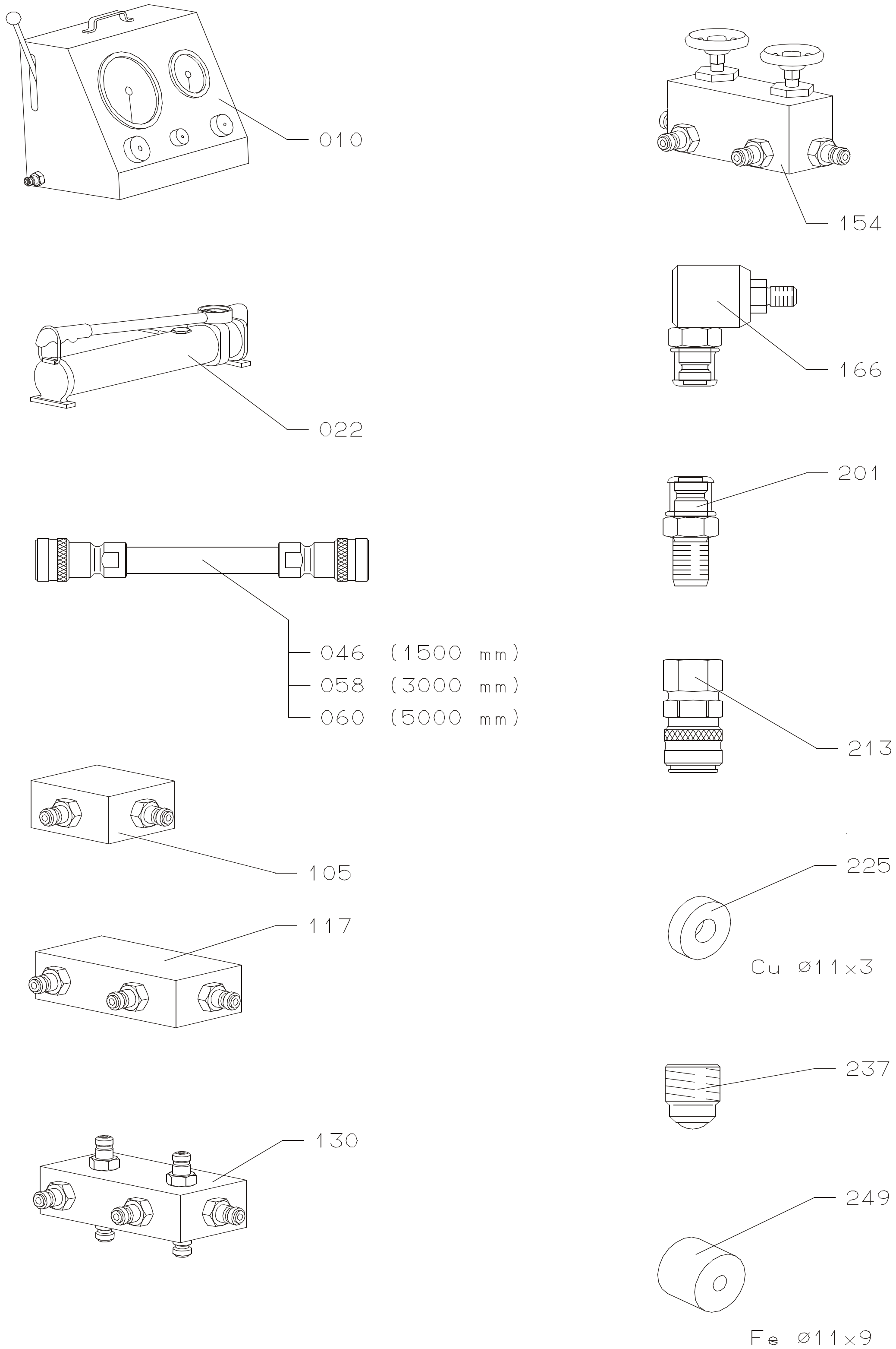
- Allow 5 minutes for the cleaning fluid to evaporate.

- With a clean leathercloth, and using circular movements, rub a mixture of fine-grained particles of Molybdenum Disulphide (MoS_2) and mineral oil (e.g. Molycote G-n Plus, or the like) hard onto the metal surface.

- Remove any excessive paste and ensure that the metal surface is only coated with a thin, uniform, layer of the above mixture.

- Protect the wet paste and cloth from dust or other foreign particles.

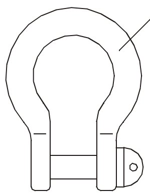
Working Pressure 1500 bar



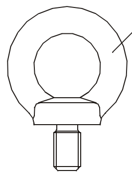
Accessories for Hydraulic Tools

Item No.	Item Description
010	Hydralic pump, pneumatic operated
022	Hydralic pump, hand operated
046	Hose with unions (1500mm), complete
058	Hose with unions (3000mm), complete
060	Hose with unions (5000mm), complete
105	3-way distributor block, complete
117	5-way distributor block, complete
130	9-way distributor block, complete
154	Conical valve, 0 - 15
166	Angle union
201	Quick coupling, male
213	Quick coupling, female
225	Disc, round-plain, Cu ø11x3mm
237	Bleeder screw
249	Disc, round-plain, Cu ø11x9mm

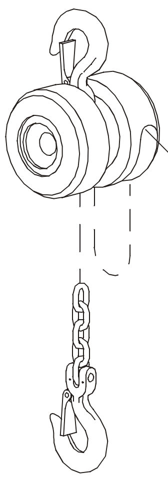
Item No.	Item Description



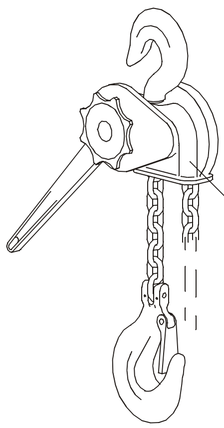
Item	Size
020	M10
031	M12
043	M16
055	M20
067	M24
079	M30



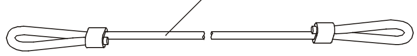
Item	Size
114	M10
126	M12
138	M16
140	M20
151	M24
163	M30



Item	Max.Load (kg)
210	500
222	1000
234	2000
258	3000



Item	Max.Load (kg)
295	3000

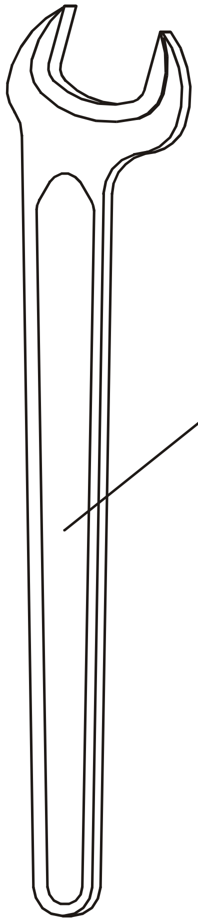


Item	Lenght (m)
305	6.0

Lifting Tools, Etc.

Item No.	Item Description
020	Forged-bent screw shackle, 10mm
031	Forged-bent screw shackle, 12mm
043	Forged-bent screw shackle, 16mm
055	Forged-bent screw shackle, 20mm
067	Forged-bent screw shackle, 24mm
079	Forged-bent screw shackle, 30mm
114	Lifting eye bolts, 10mm
126	Lifting eye bolts, 12mm
138	Lifting eye bolts, 16mm
140	Lifting eye bolts, 20mm
151	Lifting eye bolts, 24mm
163	Lifting eye bolts, 30mm
210	Chain tackle, 500kg
222	Chain tackle, 1000kg
234	Chain tackle, 2000kg
258	Chain tackle, 3000kg
295	Pull lift, 3000kg
305	Wire rope, 6m

Item No.	Item Description

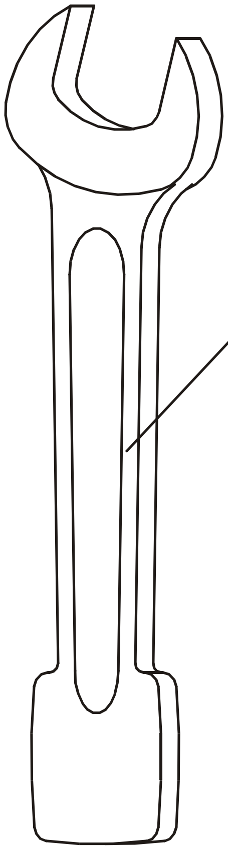


Item	Size (mm)
013	65
025	**70**
037	**75**
050	85

Open Ended Spanners

Item No.	Item Description
013	Open-ended spanner, size NV65mm
025	Open-ended spanner, size NV70mm
037	Open-ended spanner, size NV75mm
050	Open-ended spanner, size NV85mm

Item No.	Item Description



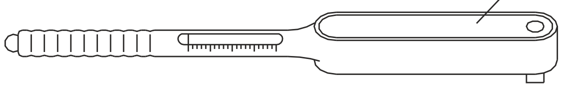
Item	Size (mm)
127	65

Open-Ended Slugging Spanners

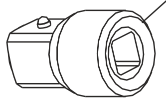
Item No.	Item Description
127	Open-ended slugging spanner, size NV65mm

Item No.	Item Description

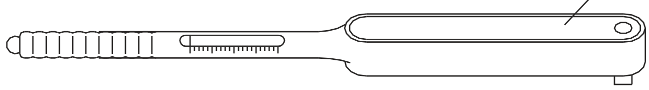
Item	Torque (Nm)
014	10-60



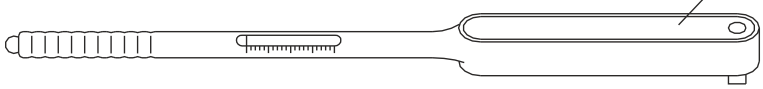
Item	Size (mm)
026	12.5/10



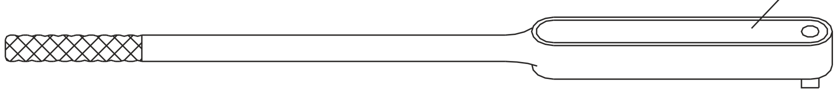
Item	Torque (Nm)
038	40-200



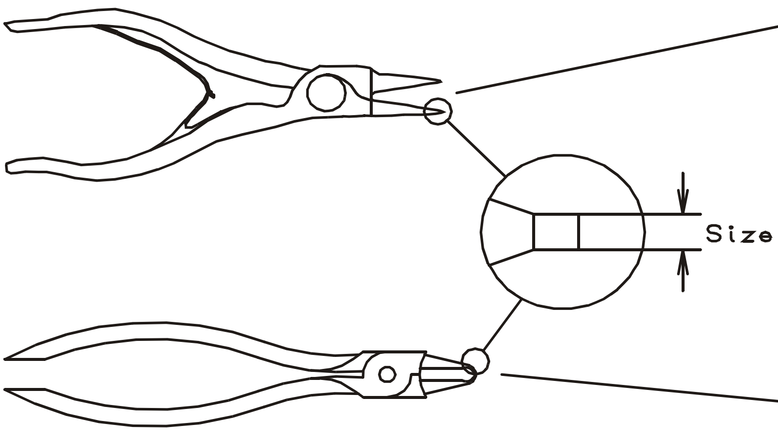
Item	Torque (Nm)
040	160-800



Item	Torque (Nm)
051	500-2000



Item No.	Item Description		
014	Torque wrench, 10-60Nm		
026	Adapter for socket wrench, 10 x 12.5 mm		
038	Torque wrench, 40 - 200Nm		
040	Torque wrench, 160 - 800Nm		
051	Torque wrench, 500 - 2000Nm		



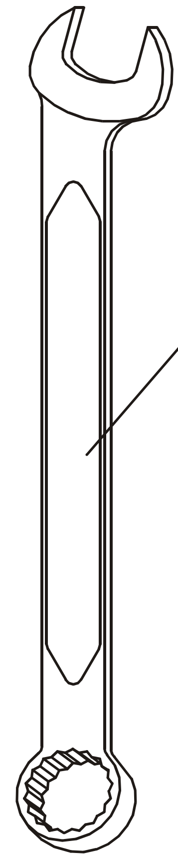
Item	Size (mm)
010	0.9
021	1.3
033	1.8

Item	Size (mm)
128	1.1
130	1.8
141	2.3
153	3.2

Pliers

Item No.	Item Description
010	Pliers for retaining ring, size 0.9mm
021	Pliers for retaining ring, size 1.3mm
033	Pliers for retaining ring, size 1.8mm
128	Pliers for retaining ring, size 1.1mm
130	Pliers for retaining ring, size 1.8mm
141	Pliers for retaining ring, size 2.3mm
153	Pliers for retaining ring, size 3.2mm

Item No.	Item Description



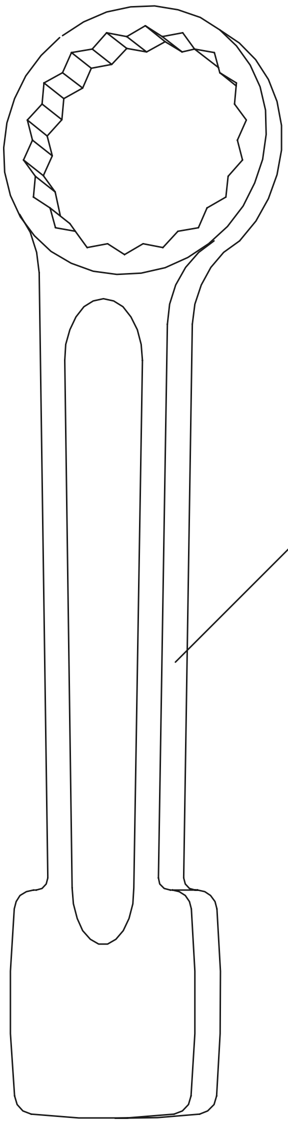
Item	Size (mm)
015	10
027	11
039	12
040	13
052	14
064	15
076	16
088	17
090	18
100	19
111	21
123	22
135	24
147	27
159	30
160	32
172	34
184	36
196	41
206	46
218	50
220	55
231	60

Combination Spanners

Item No.	Item Description
015	Combination spanners, size NV10mm
027	Combination spanners, size NV11mm
039	Combination spanners, size NV12mm
040	Combination spanners, size NV13mm
052	Combination spanners, size NV14mm
064	Combination spanners, size NV15mm
076	Combination spanners, size NV16mm
088	Combination spanners, size NV17mm
090	Combination spanners, size NV18mm
100	Combination spanners, size NV19mm
111	Combination spanners, size NV21mm
123	Combination spanners, size NV22mm
135	Combination spanners, size NV24mm
147	Combination spanners, size NV27mm
159	Combination spanners, size NV30mm
160	Combination spanners, size NV32mm
172	Combination spanners, size NV34mm
184	Combination spanners, size NV36mm
196	Combination spanners, size NV41mm
206	Combination spanners, size NV46mm
218	Combination spanners, size NV50mm
220	Combination spanners, size NV55mm
231	Combination spanners, size NV60mm

Item No.	Item Description

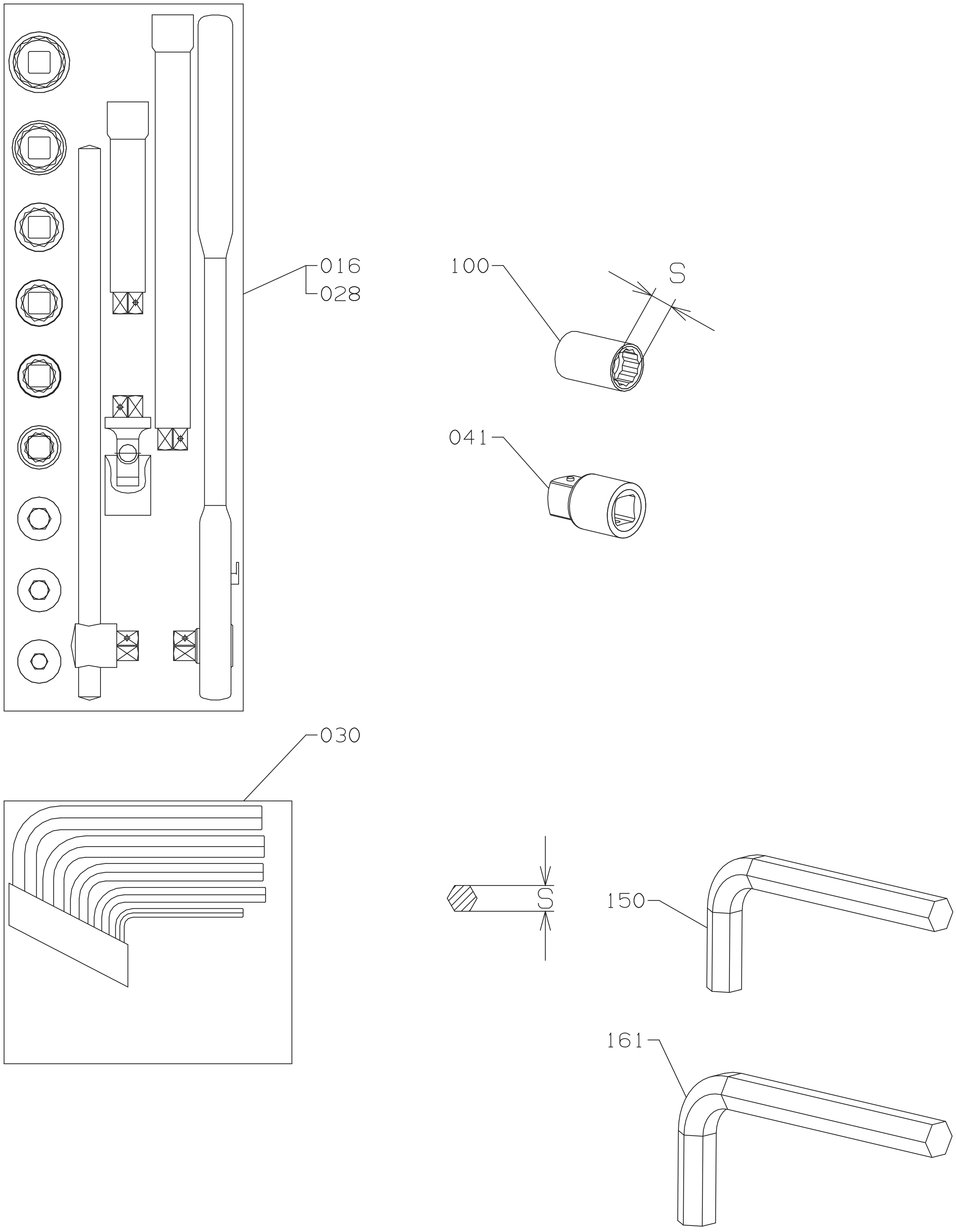
Item	Size (mm)
022	30
034	32
046	34
058	36
060	41
071	46
083	50
095	55
105	60
117	65
129	70
130	75
142	80



Ring Slugging Spanner

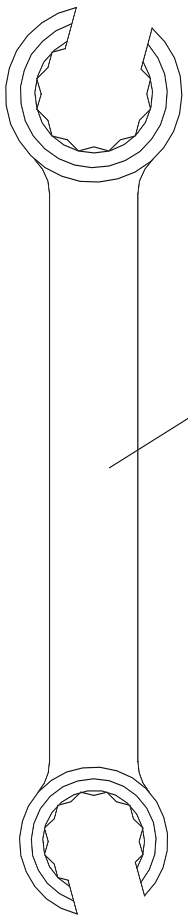
Item No.	Item Description
022	Slugging spanners, ring size NV30mm
034	Slugging spanners, ring size NV32mm
046	Slugging spanners, ring size NV34mm
058	Slugging spanners, ring size NV36mm
060	Slugging spanners, ring size NV41mm
071	Slugging spanners, ring size NV46mm
083	Slugging spanners, ring size NV50mm
095	Slugging spanners, ring size NV55mm
105	Slugging spanners, ring size NV60mm
117	Slugging spanners, ring size NV65mm
129	Slugging spanners, ring size NV70mm
130	Slugging spanners, ring size NV75mm
142	Slugging spanners, ring size NV80mm

Item No.	Item Description


016
028
100
S
041
030
S
150
161

Item No.	Item Description		
016	Tool set, complete, size 10-22 mm		
028	Tool set, complete, size 24-46 mm		
030	Hex key tool set, complete		
100	Socket spanner, 46 mm		
150	Hex key, 22 mm		
161	Hex key, 24 mm		
	Note:		
	*		

Open-ended Spanners

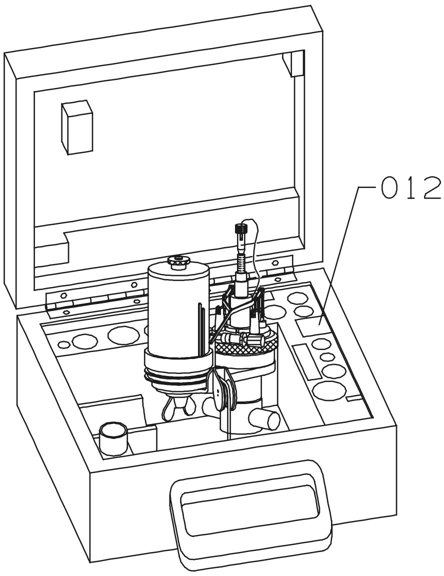


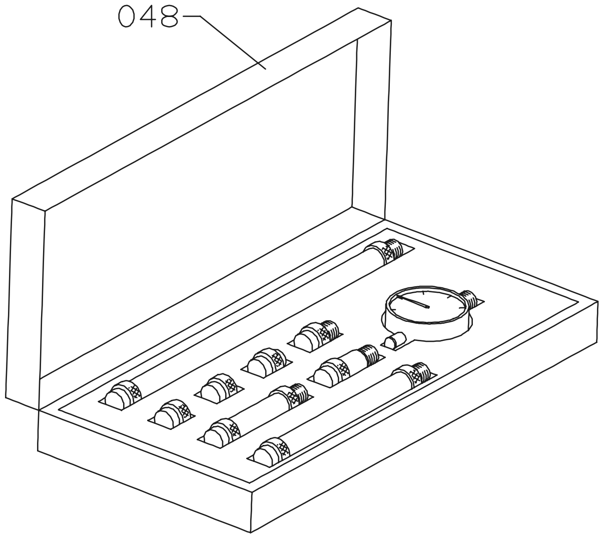
Item	Size (mm)
011	14/17
023	19/22

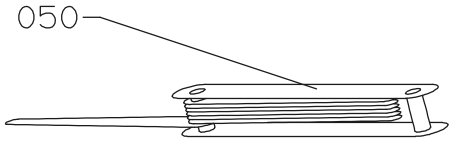
Open-ended Spanners

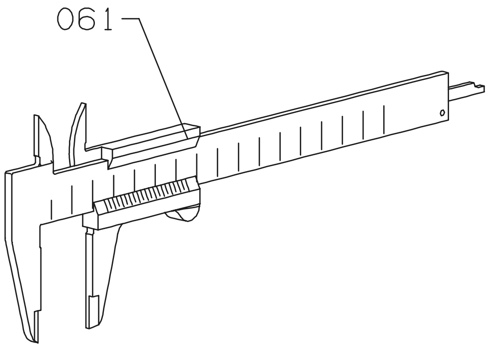
Item No.	Item Description
011	Open ring wrench, 14 - 17mm
023	Open ring wrench, 19 - 22mm

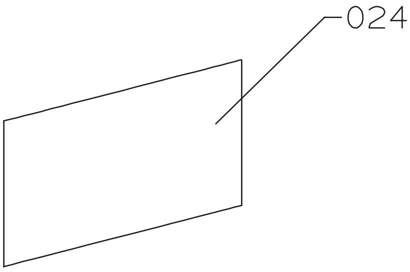
Item No.	Item Description

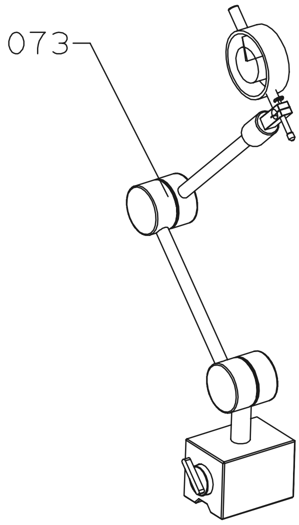

012


048


050


061

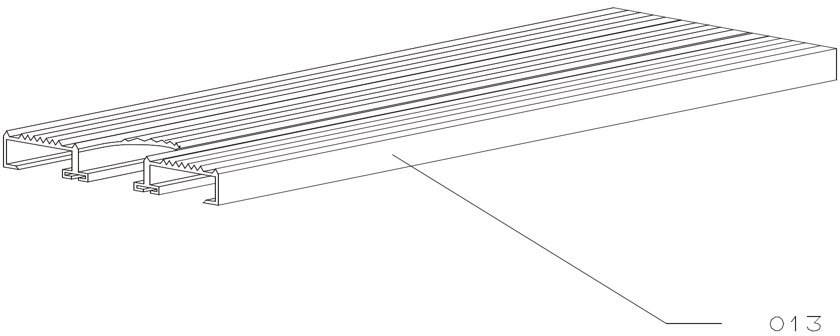

024


073

Instruments

Item No.	Item Description
012	Indicator, complete
024	Indicator paper
036	Plainmeter, complete
048	Autolog, measuring tool for crankshaft
050	Feeler gauge
061	Slide caliper
073	Dial gauge and stand tool*)
	*)To be supplied by exh. grinding mashin maker

Item No.	Item Description


013

Working Platforms

Item No.	Item Description
013	Foot grating, 800 x 250 x 40mm

Item No.	Item Description